TITLE:SMART TASK COORDINATION SYSTEM

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Problem Statement

- Many organizations assign tasks manually, which often leads to unfair workload distribution and inefficient use of employee skills
- Existing systems do not provide real-time updates on task progress, making it difficult for managers to monitor performance and deadlines
- Teams often face miscommunication, duplicated efforts, or missed responsibilities due to the absence of a centralized coordination system
- Without automated reminders and intelligent scheduling, tasks are frequently delayed, affecting overall project completion time
- Current task management tools lack smart features that analyze skills, workload, and availability before assigning tasks

Literature & Repository Analysis

Regression-Based Prediction System

Strengths

- Simple and easy to implement for predicting task delay or completion time.
- Requires low computation and provides fast predictions
- Gives quick numerical estimation of task risk or performance
- **Limitations**
- Produces only numerical outputs without detailed insights
- Cannot function alone as a fully interactive real-time web system
- Accuracy depends heavily on the quality and quantity of historical task data

Machine Learning Pipeline System

Strengths

- Improves task delay prediction accuracy using processed historical data
- Uses feature engineering to enhance model performance
- Provides graphical visualization for better decision-making
- **Limitations**
- Works mainly as an offline prediction tool
- Does not include advanced intelligent interaction like chatbots
- Requires full frontend–backend integration for real-time implementation

Proposed System

- The system will automatically assign tasks to team members based on their skills, availability, and current workload to ensure balanced distribution
- It will provide live updates on task status (Pending, In Progress, Completed) to help managers monitor progress effectively
- The system will offer a shared platform where users can comment, share files, and collaborate on tasks to reduce miscommunication
- Automated reminders and alerts will be sent for approaching deadlines, overdue tasks, and important updates.
- The system will generate reports and dashboards showing productivity, task completion rates, and workload distribution
- Different access levels (Admin, Manager, Team Member) will ensure secure and organized management of tasks and projects

Model Setup & Connectivity

AI Model Integration

- The system analyzes task requirements, user skills, and availability to determine the most suitable assignment
- It dynamically reallocates tasks when workload or priorities change to maintain balance
- The system monitors progress patterns to suggest deadline adjustments and improve overall efficiency

Why We Selected This Approach

- Automates task assignment efficiently
- Balances workload among team members
- Improves time management and productivity.
- Provides real-time tracking and performance analysis

Connectivity Verification Process

Steps Performed

- Verified network connection between client and server using ping test
- Checked server availability and ensured required services were running
- Tested database connectivity and response time
- Sent and received sample data to confirm successful communication
- **Result**
- Connection between client, server, and database was successfully established
- System confirmed reliable data transmission without errors

System Architecture

- **Users** – Stores all system users including **Admins, Project Managers, and Team Members** along with login credentials and role-based access roles.
- • **TeamMembers** – Stores team member profile mapping, including assigned projects, skill sets, workload status, and reporting manager linkage.
- •**Managers** – Stores manager details, department information, specialization, and mapped team member relationships.
- • **Tasks** – Stores task details including title, description, assigned member, priority level, deadline, status, and smart priority score.
- • **WorkloadAnalytics** – Stores performance metrics such as number of assigned tasks, completion rate, delay frequency, and predicted risk level for each team member.
- • **ExtraTasks** – Manages urgent or delayed tasks reassigned to high-priority status based on deadline risk or workload imbalance.
- • **Recommendations**– Stores system-generated task reassignment suggestions, workload balancing recommendations, and deadline extension alerts
- • **Feedback** *(optional)* –Stores team member feedback on task difficulty, collaboration quality, manager evaluation, and project performance reviews.

Database Schema

Users – Stores all system users including **Admins, Project Managers, and Team Members** along with login credentials and role-based access permissions.

• **TeamMembers** – Stores team member profile mapping, including assigned manager, project allocation, skill set, and workload linkage

• **Managers** – Stores manager details, department information, specialization, and mapped team member relationships.

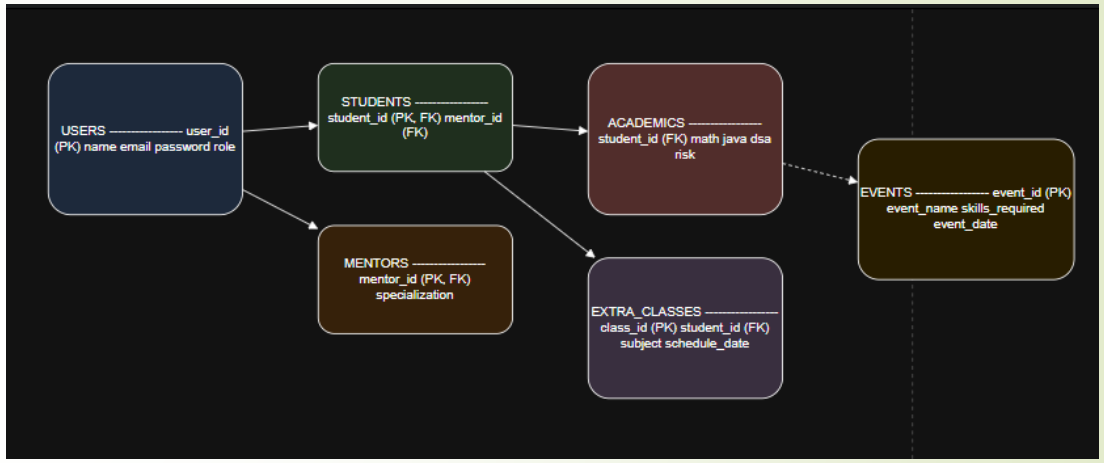
• **Tasks** – Stores task-wise details including assigned member, priority level, deadline, performance metrics, completion status, and predicted delay risk level..

• **ExtraTasks** – Manages urgent task schedules assigned to high-risk or overloaded team members based on missed deadlines or weak performance areas..

• **Projects** – Stores project details, milestones, timelines, and datasets used for performance tracking and smart task coordination.

• **Recommendations (optional/log table)** – Stores recommended task reassignments or project allocations mapped to high-performing or available team members

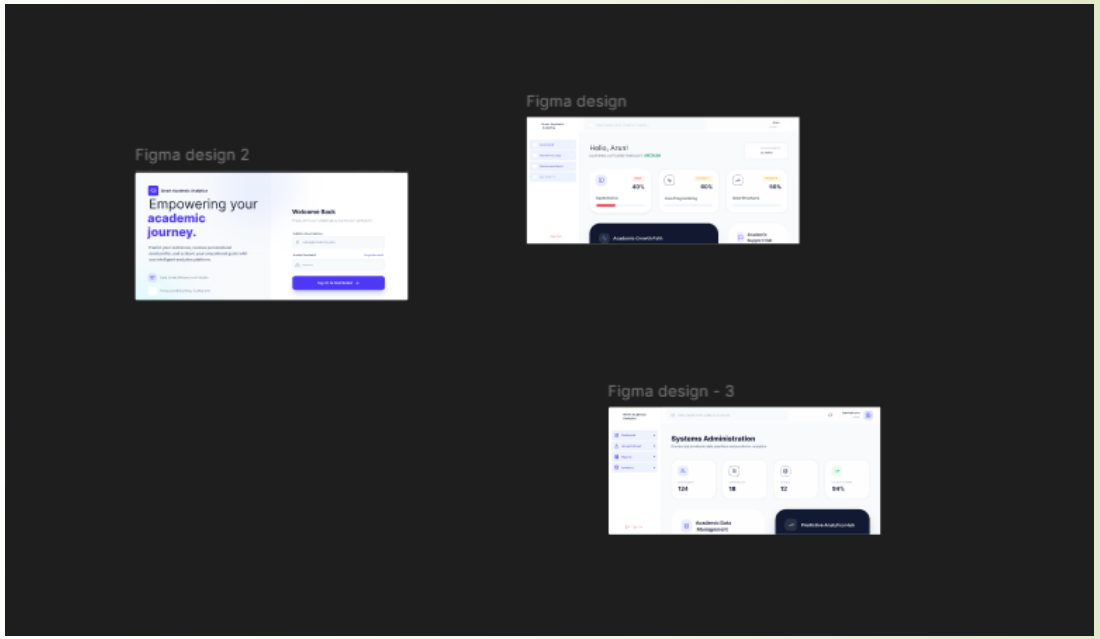
• **Feedback (optional)** – Stores team member feedback on task difficulty, collaboration quality, manager evaluation, and coordination effectiveness..



UI/UX Wireframes

Screens

- Login Page
- Admin Dashboard
- Manager Dashboard
- Team Member Dashboard
- Create Task Page
- Project View Page
- Analytics Page



Git & Environment Setup

Version Control & Secure Configuration

Setup Practices

- Git is used for version control and project management
- Node.js and Express are used to set up the backend server
- MongoDB is used as the database for storing system data
- React.js is used to develop the frontend user interface
- Environment variables (.env) are used for secure configuration management

THANK YOU